

on the LEDs powered by transistor T2, which becomes conductive at the end of one minute through NAND gates N2 and N3.

The preset VR1 associated with the audio amplifier circuit built around IC1 has to be adjusted to get clear heart beat signals. The condenser microphone should be placed on the

pulsating vein near the wrist. After the condenser microphone is fastened with a suitable belt or rubber band, you can take a fresh reading. For this you have to press the RESET/START switch S1 momentarily.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Fix the 2-pin connector

on front side of the cabinet in such a way that condenser microphone can be placed firmly on the skin of the lower wrist to get a good signal, preferably where the pulse can be easily sensed. Keep the 6V battery inside the cabinet. Fix the display LEDs in BCD format to read the pulse rate easily. ●

HOME APPLIANCE CONTROL USING TV REMOTE

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This circuit is designed to switch on/off any home or industrial appliance by using the TV/DVD remote controller. The circuit can be operated up to a distance of 5-10 metre depending on the remote used.

The circuit consists of a step-down transformer X1 (6V-0-6V, 250mA secondary), 5V regulator 7805 (IC1), two 5V, 1 change-over (C/O) relay, a timer NE555 IC (IC2), an IR receiver module (IRX1 TSOP1738) and some discrete components. The circuit works on regulated 5V, which is derived from X1 and regulated by IC1. Home appliance is controlled either by pressing any key on the re-

mo or by manually pressing switch S1 to 'on' state.

The TV/DVD remote controller produces 38kHz frequency. The IR receiver module operates at this frequency. It is used to control relay RL2. The relay triggers IC2, which is wired in a bistable mode to control the home appliance connected at the contacts of relay RL1.

Timer IC2 toggles relay RL1 when switch S1 is pressed momentarily. Threshold and trigger input pins 6 and 2 of IC2 are held at one-half of the power supply voltage (5V) by resistors R2 and R3. When output pin 3 of IC2 is high, capacitor C4 charges through resistor R4, and discharges when the output pin 3 is low. When switch S1 is

pressed, capacitor C4 voltage is applied to pins 2 and 6 of IC2, which causes the output of IC2 to change from low to high, or high to low. When switch S1 is released capacitor C4 charges or discharges to the original level at the output pin 3 of IC2.

At normal condition, when IR rays are not incident on TSOP1738, its output at pin 3 remains high. When any TV remote key is pressed, IR rays fall on the TSOP1738 and its output goes low. At the same time relay RL2 energises for a few seconds through pnp transistor T2 (BC558).

The working of the circuit is simple. Initially, when there are no IR rays falling on the IR receiver module, its

